

3S25: Software
Setup

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IC/DAQ Intro
Software Setup

SAGI Summer School 2025: IC/DAQ Intro and Software Setup

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IC/DAQ Intro

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- ❶ Instrument Control + Data Acquisition = IC/DAQ
- ❷ Instrument Control: computer automation of measurements
- ❸ Data Acquisition: Converting photons into electrical signals, storing electrical signals as bits on a computer, using bits to make progress in science.

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- ❶ Let's consider an example astronomical observation of the 21 cm neutral hydrogen line.
- ❷ In the 1s ground state of the hydrogen atom, the electron spin has a preferred (lower energy) orientation arising from it's interaction with the magnetic moment of the nucleus.

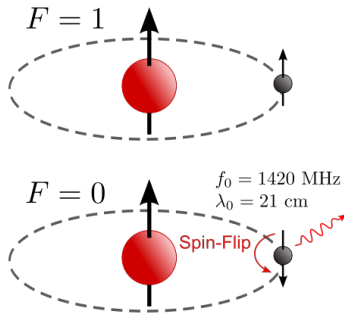


Figure: credit: Wikipedia

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- ① There is a finite energy difference between the two possible orientations of the electron spin. The electron can transition from the higher energy configuration to the lower energy configuration.
- ② To conserve energy, a photon is emitted. One of these transitions occurs at a photon frequency of $f = 1.42$ GHz, which in vacuum corresponds to a wavelength:

$$\lambda = \frac{c}{f} = \frac{299792458 \frac{\text{m}}{\text{s}}}{1.42 \cdot 10^9 \text{ Hz}} = 21 \text{ cm}. \quad (1)$$

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- 1 The 21 cm line is a very important observable in cosmology and radio astronomy. Let's turn it into bits on our computer!

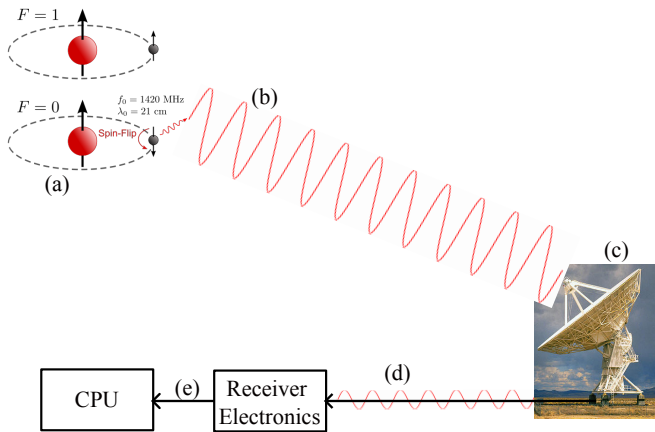


Figure: Simplified schematic of a 21 cm observation.

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- ① The basic picture of a 21 cm observation is as follows:
 - a. Spin-flip transition in neutral hydrogen in our galaxy produces electromagnetic radiation at a frequency of $f \approx 1.42$ GHz.
 - b. The radiation travels to earth.
 - c. We observe the radiation with a dish or horn antenna.
 - d. The electromagnetic signal received by the dish or antenna is coupled to transmission lines into our receiver electronics.
 - e. The receiver electronics *demodulates* and *samples* the transmission line signal so that we can store it on our computer for scientific analysis.
- ② When we say "IC/DAQ" we mean parts C-E of the detection chain.
- ③ Note that this picture is ubiquitous throughout astronomy. Replace the 21 cm line with a different observable, say the CMB or optical wavelengths, replace the dish/horn antenna with optics specific to the wavelength/frequency band of interest.

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- ❶ Throughout the course, we will be using Python to run code, make plots, and analyze data.
- ❷ Python is an **interpreted, scripting** language.
- ❸ Interpreted languages:
 - ❶ Assembly language runs directly on the CPU.
 - ❷ Compiled languages like C, are one level of abstraction above assembly. A **compiler** is a computer program that parses your C code and converts it into assembly.
 - ❸ At one level of abstraction higher, we have interpreted languages like Python. The Python interpreter is a C program which parses your python script and compiles code **on the fly**.
- ❹ In this session, we will set up a python interpreter on your computer and we will install some basic packages that we will use throughout the course.
- ❺ Another way to run python is with Google CoLab. Google provides internet access to Python interpreters that you can use without installing anything on your computer.

Software Setup: Anaconda

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- ❶ Anaconda provides a python interpreter and built-in package manager to install various essential libraries.
- ❷ Navigate to the Anaconda installer page at the following link:
<https://www.anaconda.com/download/success>
- ❸ Use the "Distribution Installers" links on the left for your operating system.

Download Now

Download Anaconda Distribution or Miniconda by choosing the proper installer for your machine. Learn the difference from our [Documentation](#).

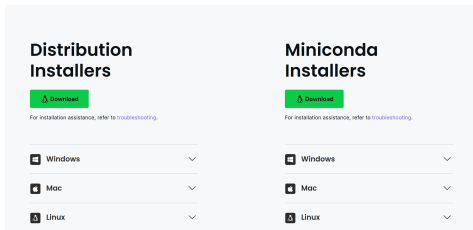


Figure: Anaconda installer page

Software Setup: Anaconda

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- ❶ If you are running Windows or macOS, execute the graphical installer and follow the instructions. If you are running Linux, execute the following commands in a terminal:
 - ❶ `cd ~/Downloads`
 - ❷ `sudo chmod +x <Anaconda installer filename>.sh`
 - ❸ `sudo ./<Anaconda installer filename>.sh`
- ❷ Verify your installation for your operating system as follows:
 - ❶ **Windows:** Search "Anaconda Prompt" in the taskbar and run this application. Run the command "conda list" in the terminal window that appears.
 - ❷ **macOS and Linux:** Open a terminal application and run "conda list".
- ❸ A successful installation will then print a list of all of the default installed python packages.

Software Setup: Anaconda

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- ➊ Now, in an Anaconda Prompt on windows or in a regular terminal on Linux and macOS, run the following command:
 - ➊ `conda create --name 3s25 python numpy scipy matplotlib pandas jupyterlab`
- ➋ This installs a python interpreter and several essential packages that we will use throughout the school.

Software Setup: Git

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- ❶ The official SAGI summer school lecture material and code is hosted on GitHub. We will use Git to "clone" this repository to our computers.
- ❷ Generally speaking, Git is an incredibly useful software tool for software version control. Version control is outside the scope of this school, but it is a very helpful skill so it is a good idea to have Git installed on your machine.
- ❸ Follow the link below to navigate to the git installer page:
<https://git-scm.com/downloads>

Software Setup: Git

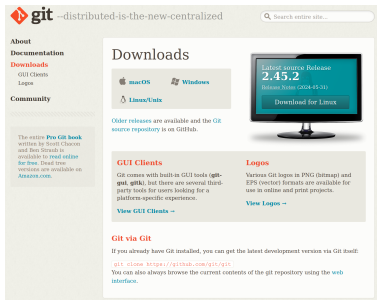
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- 1 A screenshot of the git installer page is shown below. Follow the installer instructions for your operating system and ask us if you need help.



- 2 Verify your installation by opening the "Git Bash" application on Windows or a terminal in Linux and Linux and run the command:
 - 1 `git --version`A string like "2.X.X" should appear.

Software Setup: Cloning the 3S25 Repository

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- ❶ We are collecting all of the lecture material on a GitHub page hosted by John.
- ❷ Run the following command in a terminal (linux/macOS) or the Git Bash application (Windows):
 - ❶ `git clone https://github.com/johnkimdinh/sagi_summer_school_2025.git`
- ❸ After this completes, open the "Anaconda Prompt" application and run:
 - ❶ `jupyter-lab`
- ❹ This will open up a graphical interface. Use the file browser on the left side of the interface to navigate to the cloned repository. Then open the jupyter notebook at the file path:
 - ❶ `<path to cloned repository>/lessons/ICDAQ/FourierAnalysis_PythonFFT/FourierAnalysis_Notebook0.ipynb`

Software Setup: Running the notebook

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- ❶ The notebook file that we opened contains modular blocks of python code. To execute a block, press **shift + enter** on your keyboard.
- ❷ Execute all of the cells in the notebook. If you encounter any errors, let us know and we will help you troubleshoot.